

x_1	x_2	y
10	100	5
20	200	10
30	300	15

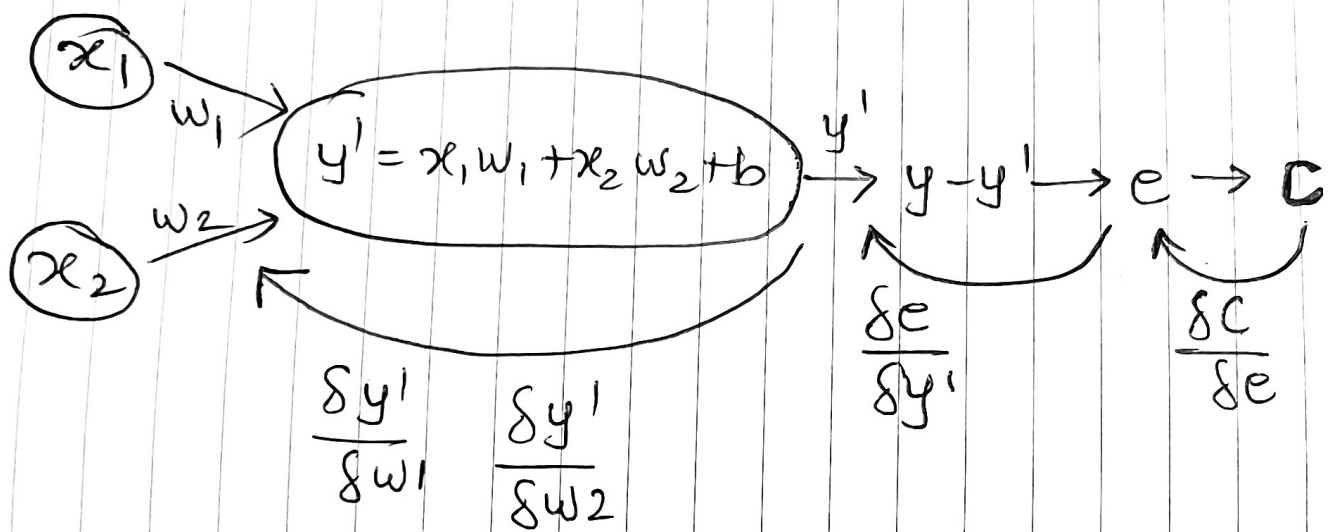
$x_1 = 10$
 $w_1 = 1$
 $x_2 = 100$
 $w_2 = 1$

$10x_1 + 100x_2 + 1$
 $y' = x_1 w_1 + x_2 w_2 + b \rightarrow y' = 111$
 $y - y' = 106 = \text{Error} = e_1$
 e_1^2

$\Rightarrow \frac{e_1^2 + e_2^2 + e_3^2}{3} = \text{MSE}$

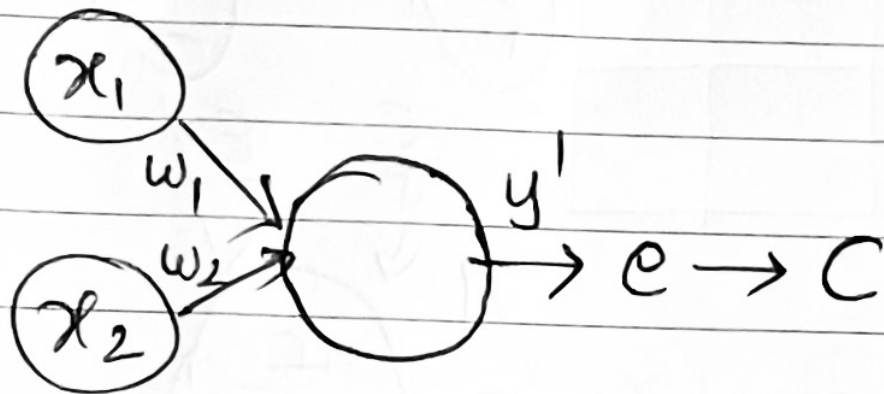
$$w_1 = w_1 - \text{LR} * \text{Pd}(w_1)$$

$$w_2 = w_2 - \text{LR} * \text{Pd}(w_2)$$



$$\frac{\partial C}{\partial w_1} = \frac{\partial C}{\partial e} * \frac{\partial e}{\partial y'} * \frac{\partial y'}{\partial w_1}$$

VGP



$$w_1 = w_1 - LR * Pd(w_1)$$

$$\frac{\delta C}{\delta w_1} = \frac{\delta C}{\delta e} * \frac{\delta e}{\delta y'} * \frac{\delta y'}{\delta w_1}$$

Gradient

$$\rightarrow 0.005$$

$$\begin{aligned} w_1 &= w_1 - (0.01) * 0.005 \\ &= w_1 - 0.00005 \end{aligned}$$

won't change much

Date: _____

EGP

$$\frac{\delta C}{\delta w_1} = 100 * 200 * 300$$

$$= 6000000$$

$$w_1 = w_1 - (0.01) * 6000000$$

$$= w_1 - 60000$$

huge change.

